

EdgeScholar: Private On-Device AI Study Copilot

Track: On-Device AI Optimized for Snapdragon-Powered HP PCs

- Core Mission: Bring high-performance, verifiable AI learning directly to student PCs.
- 100% Offline-First Architecture: Documents and personal notes never leave the local device.
- Snapdragon NPU Accelerated: Leverages Qualcomm Hexagon NPU via Qualcomm AI Hub models.
- Participant Solely Owned: MIT Licensed, independent student project built from scratch.

★ Target Hardware: HP OmniBook X & HP OmniBook Ultra (Snapdragon® X Elite / Plus)

The Student Cloud Dilemma

Why cloud-based AI tools (ChatGPT, Claude, NotebookLM) fail students daily:

- Privacy & IP Exposure: Uploading proprietary syllabi, research drafts, and exam notes risks leakage.
- Connectivity Dead Zones: Library basements, campus Wi-Fi throttles, and transit break cloud workflows.
- Cost Exclusion: \$20+/month subscription paywalls exclude millions of students worldwide.
- Ungrounded Hallucinations: Generic chatbots fabricate answers without verifiable page citations.

★ **Students need guaranteed privacy, zero recurring fees, and 100% offline reliability.**

The EdgeScholar Solution

A humanistic, distraction-free Study Desk studio running locally on Snapdragon PCs:

- Zero Cloud Leakage: Ingestion, embeddings, vector indexing, and generation run 100% on-device.
- Verifiable Margin Citations: Every answer references the exact source document filename and page.
- All-in-One Academic Studio: Natural dialogue, active recall flashcards, self-quizzes, and chapter notes.
- Real-Time Lecture Audio: Local 16kHz WAV recording and Whisper speech transcription.

★ Private • Local • Instant • Powered by Snapdragon Hexagon NPU

System Architecture & Local Pipeline

Engineered with a modular, production-ready on-device RAG stack:

- Native Desktop UI: Built in PySide6 (Qt6) with asynchronous thread workers for 60 FPS responsiveness.
- Page-Preserving Parser: PyMuPDF extracts text while strictly tracking physical book page numbers.
- Overlapping Token Chunker: 400-token sliding window with SHA-256 metadata hash integrity.
- FAISS Vector Store: Dense 384D IndexFlatIP executing inner-product cosine similarity search in
- Prompt Injection Defense: Delimiter quarantine fencing isolates textbook text from instructions.

★ **Multi-Provider Engine: Native Qualcomm QNN, ONNX Runtime, llama.cpp, and Mock fallbacks.**

Qualcomm AI Hub & Snapdragon Optimization

Natively workflows with curated models from Qualcomm AI Hub (aihub.qualcomm.com):

- Llama 3.2 3B Instruct: w4a16 INT4 quantized for Qualcomm QNN / ONNX Runtime NPU execution.
- Whisper Base English: Compiled for Qualcomm Hexagon NPU for zero-latency offline speech-to-text.
- All-MiniLM-L6-v2: Dense embeddings optimized for Hexagon NPU & Snapdragon ARM64 CPU.
- Honest Hardware Engine: Conservative status reporting (Verified vs Detected vs Emulated) with zero fake TOPS.

★ Sub-millisecond semantic search with 45 TOPS Hexagon NPU power-efficient inference.

Interactive Study Desk Features

Tactile, editorial interface designed to maximize active learning and student recall:

- 'My Shelf' Library: Instant access to multi-hundred page textbooks, papers, and lecture notes.
- Page-Citing Dialogue: Interactive badges [Sample_OS.txt, Page 2] jump directly to source context.
- Interactive Self-Quiz Studio: Generates 4-choice questions with answer keys and pedagogical rationales.
- Spaced Repetition Flashcards: Digital flip cards with front prompt and back answer for rapid exam drill.
- Structured Chapter Summaries: Automated synthesis of Key Concepts, Formulas, and Exam Takeaways.

★ **Humanistic 'Anti-AI' UI inspired by Bear & Craft — zero robotic clutter, pure focus.**

Privacy Architecture & Security Boundaries

Defense-in-depth protection ensuring absolute student privacy:

- Enforced NetworkGuard: Software-level interception strictly gating and blocking all outbound web traffic.
- Local SQLite Persistence: Encapsulates document chunks, question metrics, and session states locally.
- File Security & Boundary Checking: Rejects path traversal attempts and enforces
- One-Click Full Purge: Complete deletion manager shreds all local vector indexes and cached data on demand.

★ Zero Telemetry • Zero Cloud Logged Tokens • Complete Student Data Ownership

Deployment, Accessibility & Judging Readiness

Frictionless setup built for Snapdragon-powered HP PCs with cross-platform fallback:

- Target Deployment: Windows 11 ARM64 on HP OmniBook X / HP OmniBook Ultra.
- Universal Judging Compatibility: Auto-detects macOS, Linux, and Windows to guarantee flawless evaluation.
- Pre-Seeded Academic Fixtures: Comes pre-loaded with Operating Systems, Discrete Math, and Python notes.
- Single Command Execution: 'python -m app.main' runs out of the box with zero cloud API keys needed.

★ Ready for evaluators to test immediately without configuring keys or uploading files.

Rigorous Verification & Test Results

Every subsystem and functional pipeline has been audited and validated:

- 20 / 20 Subsystems Verified: End-to-end audit passed cleanly (scripts/verify_all_features.py).
- 31 / 31 Unit & Integration Tests: 100% green test suite in pytest executing in 4.19 seconds.
- 17 / 17 Installation Dependencies: PySide6, PyMuPDF, FAISS, sentence-transformers all validated.
- Real Hardware Instrumentation: High-resolution timers (time.perf_counter()) measuring genuine latency.

★ **Audit Score: 100% Verified Operational • Zero Synthesized Benchmarks**

Why EdgeScholar Wins

The definitive on-device AI copilot for Snapdragon-powered HP PCs:

- Exact Match to Guidelines: Specifically architected for Snapdragon HP PCs using Qualcomm AI Hub.
- Top Score in Tie-Breaker: Production RAG, QNN NPU abstraction, FAISS indexing, and prompt fencing.
- Tangible Student Impact: Frees students from subscriptions (\$240/yr saved) and protects research IP.
- Public Code & Live Repository: <https://github.com/sahilbisen52-lang/edge-scholar>

★ **EdgeScholar turns Snapdragon-powered HP PCs into an invincible, private study sanctuary.**